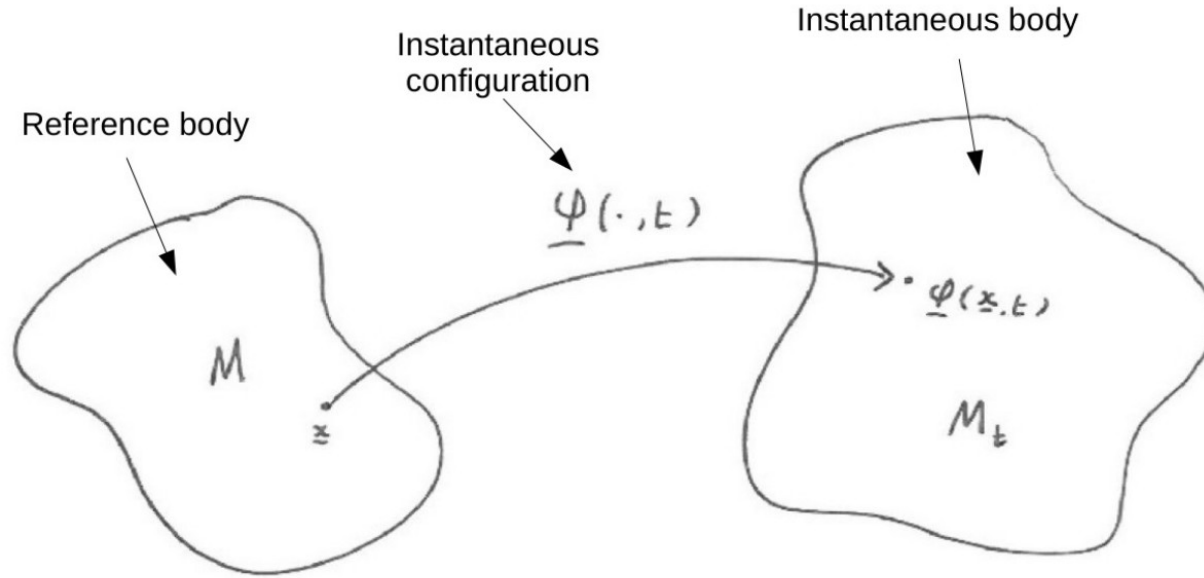


Constitutive theory

From last time



$$\mathcal{L}(\mathbf{x}, t, \varphi, \mathbf{v}, \mathbf{F}) = \frac{1}{2} \rho(\mathbf{x}) \|\mathbf{v}\|^2 - W(\mathbf{x}, \mathbf{F})$$

$$W[\mathbf{x}, \mathbf{F}(\mathbf{x}, t)] = W[\mathbf{x}, \mathbf{Q}(t) \mathbf{F}(\mathbf{x}, t)],$$

Material frame indifference

Polar decomposition theorem

For any **invertible matrix** with a **positive determinant**, we can write:

Rotation matrix

$$\mathbf{F} = \mathbf{R}\mathbf{U} = \mathbf{V}\mathbf{R}$$

Symmetric positive-definite matrices

U and **V** have the **same eigenvalues** – same magnitude of stretching about rotated principal axes

Proof is non-examinable

Rotational invariance of strain energy

We require the invariance property:

$$W(\mathbf{x}, \mathbf{QF}) = W(\mathbf{x}, \mathbf{F})$$

Writing $\mathbf{F} = \mathbf{RU}$, it follows that:

$$W(\mathbf{x}, \mathbf{F}) = W(\mathbf{x}, \mathbf{RU}) = W(\mathbf{x}, \mathbf{U}),$$

This shows that strain energy is only affected by the **stretching part** of the deformation gradient

Reduced strain energy function

We have shown that the strain energy must take the form:

$$W(\mathbf{x}, \mathbf{F}) = U(\mathbf{x}, \mathbf{C}),$$

where we have introduced the **right Cauchy-Green deformation tensor**:

$$\mathbf{C} = \mathbf{F}^T \mathbf{F}$$

Note that $\mathbf{U}$ is the square-root of $\mathbf{C}$. We use $\mathbf{C}$ in practice because its simpler

Equivalently, we could use the **finite strain tensor**:

$$\mathbf{E} = \frac{1}{2}[\mathbf{C} - \mathbf{I}]$$

Consequences of the result

Recall the first Piola-Kirchhoff stress tensor defined by:

$$T_{ij} = \frac{\partial W}{\partial F_{ij}}$$

Using the chain rule, we can then write:

$$T_{ij} = \frac{\partial U}{\partial C_{kl}} \frac{\partial C_{kl}}{\partial F_{ij}}.$$

From a careful application of index-notation

$$\frac{\partial C_{kl}}{\partial F_{ij}} = \frac{\partial}{\partial F_{ij}} (F_{mk} F_{ml}) = \delta_{im} \delta_{jk} F_{ml} + F_{mk} \delta_{im} \delta_{jl} = \delta_{jk} F_{il} + \delta_{jl} F_{ik},$$

Consequences of the result

We have now shown that:

$$T_{ij} = F_{ik} S_{kj}$$

Where we have defined the **second Piola-Kirchhoff stress tensor**:

$$S_{ij} = 2 \frac{\partial U}{\partial C_{ij}}$$

Note that **S** is a **symmetric tensor** – in the third problem set you will see that this symmetry is equivalent to the **conservation of angular momentum**

Equilibria and linearisation

Equilibrium configurations

Equations of motion and boundary conditions:

$$\rho \frac{\partial v_i}{\partial t} - \frac{\partial T_{ij}}{\partial x_j} = 0, \quad T_{ij} \hat{n}_j = 0.$$

Equilibrium conditions:

$$\varphi^0(\mathbf{x}) = \mathbf{x} \quad - \frac{\partial T_{ij}^0}{\partial x_j} = 0 \quad T_{ij}^0 \hat{n}_j = 0.$$

Stress-free equilibrium configurations are a possible solution!

Linearisation about equilibrium

We consider an initial disturbance about a stress-free equilibrium:

$$\varphi_i(\mathbf{x}, 0) = x_i + s u_i^0(\mathbf{x}) \quad v_i(\mathbf{x}, 0) = s v_i^0(\mathbf{x}).$$

Here “s” is a perturbation parameter that is used for book-keeping. The case $s=0$ corresponds to equilibrium

We expand the motion in powers of s:

$$\varphi_i(\mathbf{x}, t) = x_i + s u_i(\mathbf{x}, t) + O(s^2)$$

The first-order coefficient, $\mathbf{u}$, is known as the **displacement vector**

First order expansions

First-order expansions of the velocity and deformation gradient:

$$v_i = s \frac{\partial u_i}{\partial t} + O(s^2)$$

$$F_{ij} = \delta_{ij} + s \frac{\partial u_i}{\partial x_j} + O(s^2).$$

First order expansions

From the definition of the first Piola-Kirchhoff stress, we find:

$$T_{ij} = s A_{ijkl} \frac{\partial u_k}{\partial x_l} + O(s^2).$$

where we have defined the **elastic tensor**:

$$A_{ijkl}(\mathbf{x}) = \frac{\partial^2 W}{\partial F_{ij} \partial F_{kl}}(\mathbf{x}, \mathbf{I}),$$

This is **Hooke's law**, but its nothing more than the result of linearisation

Note the **hyperelastic symmetry** which will be important:

$$A_{ijkl} = A_{klij}.$$

Linearised equations of motion

Linearised equation of motion:

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = 0$$

Linearised boundary condition:

$$n_j A_{ijkl} \frac{\partial u_k}{\partial x_l} = 0.$$

These linearised equations are sufficient for describing most aspects of seismic wave propagation within the Earth

Additional symmetries

About a stress-free equilibrium, the elastic tensor possess additional symmetries:

$$A_{ijkl} = A_{jikl} = A_{ijlk}$$

You will show this in the third problem set

This allows us to write the **linearised first Piola-Kirchhoff** stress tensor:

$$A_{ijkl} \frac{\partial u_k}{\partial x_l} = A_{ijkl} \frac{1}{2} \left(\frac{\partial u_k}{\partial x_l} + \frac{\partial u_l}{\partial x_k} \right)$$

where on the right we see the **linearised strain tensor**, which is often called *the* strain tensor in elementary (but inaccurate) discussions of elasticity

Material symmetries

The material symmetry group

Recall the invariance property under super-imposed rotations:

$$W(\mathbf{x}, \mathbf{QF}) = W(\mathbf{x}, \mathbf{F})$$

What happens if we do the rotation first?

$$W(\mathbf{x}, \mathbf{FQ}) = W(\mathbf{x}, \mathbf{F}),$$

This condition does not hold in general for all rotation matrices, $\mathbf{Q}$

When it does, we say that $\mathbf{Q}$ is in the **material symmetry group** at this point

This reflects an invariance of the strain energy with respect to **orientation of stretching**. Physically, it reflects **lattices symmetries** in crystalline materials

Isotropic materials

An elastic body is **isotropic** at a point if the corresponding material symmetry group comprises **all rotation matrices**

Otherwise, the body is **anisotropic** at this point

Making use of material frame indifference, isotropy requires:

$$U(\mathbf{C}) = U(\mathbf{Q}^T \mathbf{C} \mathbf{Q}).$$

for all rotation matrices

It follows that the strain energy depends only on the **eigenvalues** of the right Cauchy-Green deformation tensor

Isotropic materials

It is only a matter of tedious algebra to show that:

$$A_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{kj}),$$

Specification of the full elastic tensor requires only the two **Lame parameters**

If the material is also homogeneous, the linearised equations reduce to:

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} - (\lambda + \mu) \nabla (\nabla \cdot \mathbf{u}) - \mu \nabla^2 \mathbf{u} = 0.$$

This is the form presented in **1B dynamics**, but there are a **lot of assumptions** necessary to get here, most of which **don't hold in the Earth**

Elastic fluids

In an **elastic fluid**, there is no preferred shape, only a **preferred volume**

The strain energy depends only on the Jacobian of the motion:

$$W(\mathbf{x}, \mathbf{F}) = V(\mathbf{x}, J),$$

Equivalently, the material symmetry group comprises all matrices with **determinant equal to one** (the special linear group)

In this case, the elastic tensor takes the simple form:

$$A_{ijkl} = \kappa \delta_{ij} \delta_{kl}$$